

Notes on the variational structure of minimizers for superlevel sets of logarithmic potentials

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Abstract

We study the minimization problem

$$L(\mu) := |\{x \in \mathbb{R} : U_\mu(x) > 0\}|$$

for logarithmic potentials of probability measures supported on $[-1, 1]$, equivalently the Erdős–Herzog–Piranian polynomial problem in logarithmic-potential form. We begin by recording the basic setup: existence of a minimizer, the openness and boundedness of the superlevel set, and the classical half-interval consequence of the sign of the barycenter. We then prove a strict collapse lemma showing that, for a genuine minimizer, each connected component of $\{U_\mu > 0\}$ carries a single atom of μ . Motivated by this, we discuss a normalization that places a minimizing measure into a one-sided configuration of the form $\{-1\} \cup [0, 1]$. We next set up a regular first-variation framework for the length functional: assuming C^1 regularity in neighborhoods of the free boundary and compactness of the interior active zero set, we derive a one-sided first-variation formula, identify the associated boundary measure, and carry out a dual-completion step that adds a multiplier on the interior active set. Under the additional hypothesis $\mu|_{Z_0} \ll dx$, this yields the vanishing of the dual potential U_λ for μ -almost every point. Finally, we work out the one-cut constant-potential ansatz explicitly and discuss the transport calculation, which leads to the same principal-value equations from the variational side.

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1 Setup and notation

Let μ be a nonzero finite positive Borel measure on $\mathbb{R}$ supported in $[-1, 1]$, and define its logarithmic potential by

$$U_\mu(x) := \int_{\mathbb{R}} \log \frac{1}{|x - t|} d\mu(t).$$

We use the sign convention above, so the superlevel set $\{U_\mu > 0\}$ corresponds to the set where $|P| < 1$ in the polynomial formulation. Equivalently, one may work with the opposite-sign convention

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¹<https://chatgpt.com/share/69b86cf8-f7d8-8010-b2ee-ed3b553f875d>;
<https://chatgpt.com/share/69b95ed7-ade4-8010-b783-45cf983c3798>;
<https://chatgpt.com/share/69bb0b4a-140c-8010-8979-8122891496a3>.

Some chats cannot be shared due to persistent errors and are not linked here.

$\int \log|x-t| d\mu(t)$ and the sublevel set $\{U_\mu < 0\}$ that appears in parts of the forum discussion. Set

$$E_\mu := \{x \in \mathbb{R} : U_\mu(x) > 0\}, \quad L(\mu) := |E_\mu|,$$

where $|\cdot|$ denotes Lebesgue measure.

This is the logarithmic-potential reformulation of the original polynomial problem. Indeed, if

$$P(x) = \prod_{j=1}^n (x - t_j) \quad (t_j \in [-1, 1])$$

is monic and $\mu := \frac{1}{n} \sum_{j=1}^n \delta_{t_j}$ is its empirical root measure, then

$$\log \frac{1}{|P(x)|} = nU_\mu(x),$$

so

$$\{x : |P(x)| < 1\} = \{x : U_\mu(x) > 0\} = E_\mu.$$

For every $c > 0$ one has $U_{c\mu} = cU_\mu$, hence $E_{c\mu} = E_\mu$ and $L(c\mu) = L(\mu)$. Thus one may formulate the infimum problem either for probability measures or for arbitrary nonzero finite positive measures. In this note we usually normalize to probability measures when discussing minimizers, but we freely use unnormalized positive measures for perturbative arguments. Following Tao, we write

$$L^- := \inf\{L(\mu) : \mu \text{ a probability measure supported in } [-1, 1]\}.$$

By compactness of the space of probability measures on $[-1, 1]$ in the vague topology and Tao's lower semicontinuity argument, a minimizing probability measure exists; compare [3, Lemmas 1.1 and 3.1].

By lower semicontinuity of U_μ , the set E_μ is open; because μ is nonzero and supported in the compact interval $[-1, 1]$, one also has $U_\mu(x) \rightarrow -\infty$ as $|x| \rightarrow \infty$, so E_μ is bounded. Hence E_μ is a countable union of disjoint open intervals.

We will use the following classical input. If the mean of μ is nonnegative, then $(0, 1) \subset E_\mu$, and if the mean is nonpositive, then, by reflection, $(-1, 0) \subset E_\mu$. This is the Jensen-inequality portion of [2, Theorem 1], and it is also recorded explicitly as [3, Lemma 3.2].

2 A strict collapse lemma

We begin with a strengthened barycenter-collapse statement: on a connected component of E_μ , collapsing the mass to its barycenter strictly decreases the superlevel-set length unless that piece of the measure is already a single atom.

Lemma 2.1 (Strict collapse on a positive component). *Let μ be a finite positive measure supported in $[-1, 1]$, and let $I = (a, b)$ be a connected component of E_μ . Then $\mu(I) > 0$. Let*

$$m := \mu(I), \quad c := \frac{1}{m} \int_I t d\mu(t).$$

Then $c \in I \cap [-1, 1]$, and define

$$\tilde{\mu} := \mu|_{\mathbb{R} \setminus I} + m\delta_c.$$

Then

$$L(\tilde{\mu}) \leq L(\mu),$$

with equality if and only if $\mu|_I = m\delta_c$.

Proof. If $\mu(I) = 0$, then U_μ is convex on I , because for each fixed $t \notin I$ the function $x \mapsto \log \frac{1}{|x-t|}$ is convex on I . This is impossible: $U_\mu > 0$ on I , while $U_\mu \leq 0$ on $\mathbb{R} \setminus E_\mu$, so in particular at the boundary points a and b . Hence $m > 0$.

Fix $x \notin I$. Then the function

$$t \mapsto \log \frac{1}{|x-t|}$$

is strictly convex on I , since its second derivative is $(x-t)^{-2} > 0$. Jensen's inequality therefore gives

$$\int_I \log \frac{1}{|x-t|} d\mu(t) \geq m \log \frac{1}{|x-c|},$$

with equality if and only if $\mu|_I$ is the point mass $m\delta_c$. Adding the unchanged contribution from $\mu|_{\mathbb{R} \setminus I}$, we obtain

$$U_{\tilde{\mu}}(x) \leq U_\mu(x) \quad (x \notin I),$$

and this inequality is strict for every $x \notin I$ unless $\mu|_I = m\delta_c$.

It follows immediately that

$$U_{\tilde{\mu}}(x) \leq 0 \quad (x \in \mathbb{R} \setminus E_\mu),$$

so

$$E_{\tilde{\mu}} \setminus I \subseteq E_\mu \setminus I.$$

If $\mu|_I = m\delta_c$, then $\tilde{\mu} = \mu$ and we are done. Assume henceforth that $\mu|_I$ is not a point mass.

We next look at the endpoints. Since $a, b \notin E_\mu$, neither endpoint can carry an atom of μ ; otherwise $U_\mu(a) = +\infty$ or $U_\mu(b) = +\infty$, contradicting $a, b \notin E_\mu$. Hence U_μ is continuous at a and b . Because $a, b \in \text{bd } E_\mu$ and $U_\mu > 0$ on I , we must therefore have

$$U_\mu(a) = U_\mu(b) = 0.$$

Applying the strict Jensen inequality at $x = a$ and $x = b$ yields

$$U_{\tilde{\mu}}(a) < 0, \quad U_{\tilde{\mu}}(b) < 0.$$

Again $\tilde{\mu}$ has no atom at a or b , so $U_{\tilde{\mu}}$ is continuous there. Consequently there exists $\varepsilon > 0$ such that

$$U_{\tilde{\mu}}(x) < 0 \quad \text{for } x \in (a, a + \varepsilon) \cup (b - \varepsilon, b).$$

Thus

$$E_{\tilde{\mu}} \cap I \subseteq [a + \varepsilon, b - \varepsilon],$$

which implies

$$|E_{\tilde{\mu}} \cap I| < |I|.$$

Combining this with the inclusion outside I gives

$$L(\tilde{\mu}) < L(\mu).$$

This proves the strict case, and the equality case has already been treated. $\square$

Corollary 2.2 (One atom on each positive component). *Let μ be a minimizer of $L(\mu)$ among probability measures supported in $[-1, 1]$. Then for every connected component I of E_μ there exists $x_I \in I$ such that*

$$\mu|_I = \mu(I)\delta_{x_I}.$$

In particular, I contains exactly one support point of μ .

Proof. If $\mu|_I$ were not a point mass, Lemma 2.1 would produce another admissible probability measure supported in $[-1, 1]$ with strictly smaller superlevel-set length, contradicting the minimality of μ . The final sentence follows immediately from the explicit form of $\mu|_I$. $\square$

3 Normalization up to reflection and translation

Using only the setup from Section 1 and the one-atom-per-component conclusion from Corollary 2.2, one obtains the usual support reduction without any auxiliary minimum-variance hypothesis.

Proposition 3.1 (Normalization of a minimizer). *Let μ be a minimizing probability measure for L^- . Then, after possibly reflecting $x \mapsto -x$ and then translating, one may assume that*

$$(-1, 0) \subset E_\mu, \quad \text{supp } \mu \subseteq \{-1\} \cup [0, 1], \quad \mu(\{-1\}) \geq \frac{1}{2}.$$

Proof. Reflect, if necessary, so that the mean of μ is nonpositive. By the Jensen lemma quoted in Section 1, one then has

$$(-1, 0) \subset E_\mu.$$

Let $I = (\alpha, \beta)$ be the connected component of E_μ containing $(-1, 0)$. By Corollary 2.2,

$$\mu|_I = p \delta_c$$

for some $c \in I$ and some $p = \mu(I) > 0$.

We first show that c is the leftmost support point of μ . Suppose for contradiction that there exists $t \in \text{supp } \mu$ with $t < c$. Since I contains only the atom at c , one has $t \notin I$, hence $t \leq \alpha$. Because $(-1, 0) \subset I$, necessarily $\alpha \leq -1$. Since also $t \in [-1, 1]$ and $t \leq \alpha$, this forces $t = \alpha = -1$. But then $-1 \in \text{supp } \mu$, so $U_\mu(-1) = +\infty$ and therefore $-1 \in E_\mu$, contradicting $\alpha = -1 \notin I$. Thus no support point lies to the left of c .

If $c > -1$, translate the whole measure left by $s := -1 - c \leq 0$, and continue to denote the translated measure by μ . Since c was the leftmost support point, every support point t satisfies $t \in [c, 1]$, hence after translation $t + s \in [-1, -c] \subset [-1, 1]$. Thus the translated measure is still admissible. Translation preserves the quantity $L(\mu)$ exactly, so the translated measure is again a minimizer. Its mean is the old mean plus s , and is therefore still nonpositive. By construction the distinguished atom now sits at

$$c = -1.$$

Applying again the Jensen lemma from Section 1, we still have

$$(-1, 0) \subset E_\mu.$$

Let $I = (\alpha, \beta)$ denote the connected component of E_μ containing $(-1, 0)$ after this translation. Since $-1 \in I \cap \text{supp } \mu$, Corollary 2.2 shows that

$$\mu|_I = p \delta_{-1},$$

so there is no other support point in I . Because $(-1, 0) \subset I$, its right endpoint satisfies $\beta \geq 0$. Consequently,

$$\text{supp } \mu \subseteq \{-1\} \cup [\beta, 1] \subseteq \{-1\} \cup [0, 1].$$

Write

$$p := \mu(\{-1\}).$$

We prove that $p \geq \frac{1}{2}$.

Because $\beta \in \text{bd } E_\mu$, the point β carries no atom of μ and hence $U_\mu(\beta)$ is finite; since $\beta \notin E_\mu$, one has $U_\mu(\beta) \leq 0$. Jensen's inequality gives

$$0 \leq -U_\mu(\beta) \leq \log \int |\beta - t| d\mu(t),$$

and hence

$$\int |\beta - t| d\mu(t) \geq 1.$$

If $\beta > 0$, then using $\text{supp } \mu \subseteq \{-1\} \cup [\beta, 1]$ we obtain

$$\int |\beta - t| d\mu(t) \leq (\beta + 1)p + (1 - \beta)(1 - p) = 1 - \beta + 2\beta p.$$

Therefore $1 \leq 1 - \beta + 2\beta p$, which implies $p \geq \frac{1}{2}$.

It remains to consider the case $\beta = 0$. Then the inequality above becomes

$$\int |t| d\mu(t) \geq 1.$$

Since $|t| \leq 1$ on $[-1, 1]$, equality must hold, and therefore $|t| = 1$ for μ -almost every t . As we already know that $\text{supp } \mu \subseteq \{-1\} \cup [0, 1]$, this forces

$$\mu = p\delta_{-1} + (1 - p)\delta_1.$$

The mean of μ is nonpositive, so

$$-p + (1 - p) \leq 0,$$

which is exactly $p \geq \frac{1}{2}$. □

Corollary 3.2 (The component through $(-1, 0)$ reaches $-\sqrt{2}$). *Under the normalization of Proposition 3.1, if $I = (\alpha, \beta)$ is the connected component of E_μ containing $(-1, 0)$, then*

$$(-\sqrt{2}, 0) \subset I.$$

In particular,

$$\alpha \leq -\sqrt{2}.$$

Proof. By Proposition 3.1, we may write

$$\mu = p\delta_{-1} + \nu, \quad p := \mu(\{-1\}) \geq \frac{1}{2},$$

where ν is a positive measure supported on $[0, 1]$ with total mass $1 - p$. Let $x \in (-\sqrt{2}, 0)$. For $t \in [0, 1]$ one has $|x - t| = t - x \leq 1 - x$, so

$$\log \frac{1}{|x - t|} \geq \log \frac{1}{1 - x}.$$

Therefore

$$U_\mu(x) \geq p \log \frac{1}{|x + 1|} + (1 - p) \log \frac{1}{1 - x}.$$

Since

$$\log \frac{1 - x}{|x + 1|} > 0 \quad (x < 0),$$

the right-hand side is increasing in p , and thus

$$U_\mu(x) \geq \frac{1}{2} \log \frac{1}{|x + 1|(1 - x)}.$$

If $-1 < x < 0$, then $|x + 1| = 1 + x$ and

$$|x + 1|(1 - x) = 1 - x^2 < 1,$$

so $U_\mu(x) > 0$. If $-\sqrt{2} < x \leq -1$, then $|x + 1| = -1 - x$ and

$$|x + 1|(1 - x) = x^2 - 1 < 1,$$

again giving $U_\mu(x) > 0$. Thus $(-\sqrt{2}, 0) \subset E_\mu$, and since this interval meets I on $(-1, 0)$ it must lie in the same connected component I . This proves the claim. □

4 Regular first variation and dual completion

In this section we set up a regular first-variation framework for the length functional. The hypotheses have two distinct roles. Along the moving boundary $\text{bd}(E)$ we assume C^1 regularity in neighborhoods of the boundary points, together with a nondegenerate sign change, so that the endpoints of the positivity set can be tracked under perturbation. On the interior zero set

$$Z_0 := \{U = 0\} \setminus \text{bd}(E),$$

we assume continuity, which prevents new positive components from being created under admissible perturbations.

After proving the boundary first-variation formula, we specialize to a minimizing measure μ and record a set of standing assumptions for the rest of the section. Under those assumptions we construct the boundary measure λ_∂ , then adjoin a multiplier ξ supported on Z_0 by a Hahn–Banach separation argument. The resulting dual measure $\lambda = c\lambda_\partial + \xi$ satisfies averaged positivity on $[-1, 1]$, pointwise positivity away from $Z_0 \cup \text{bd}(E_\mu)$, and the mutual-energy identity $\int U_\lambda d\mu = 0$. If in addition $\mu|_{Z_0} \ll dx$, then $U_\lambda = 0$ for μ -almost every point.

Proposition 4.1 (One-sided regular first variation). *Let $U : \mathbb{R} \rightarrow [-\infty, \infty]$ be lower semicontinuous, and suppose*

$$E := \{x : U(x) > 0\} = \bigsqcup_{j=1}^N (a_j, b_j)$$

is a finite union of disjoint bounded open intervals. Assume moreover that U is continuous on $\mathbb{R} \setminus E$, that U is C^1 on a neighborhood of each boundary point, and that

$$U(a_j) = U(b_j) = 0, \quad U'(a_j) > 0, \quad U'(b_j) < 0.$$

Let

$$Z_0 := \{U = 0\} \setminus \text{bd}(E).$$

Assume that Z_0 is compact and that $U(x) < 0$ for $|x|$ sufficiently large. Let $V : \mathbb{R} \rightarrow [-\infty, \infty]$ satisfy the following four conditions:

- (i) V is C^1 on a neighborhood of $\text{bd}(E)$ and continuous on a neighborhood of Z_0 ;
- (ii) $\sup_{\mathbb{R} \setminus E} V < \infty$;
- (iii) $V < 0$ on Z_0 ;
- (iv) V is bounded below on every compact subset of E .

Then for all sufficiently small $\varepsilon > 0$, the set

$$E_\varepsilon := \{x : U(x) + \varepsilon V(x) > 0\}$$

is again a union of N intervals $(a_j(\varepsilon), b_j(\varepsilon))$, and

$$\frac{d}{d\varepsilon} \Big|_{\varepsilon=0^+} |E_\varepsilon| = \sum_{j=1}^N \left(\frac{V(a_j)}{|U'(a_j)|} + \frac{V(b_j)}{|U'(b_j)|} \right).$$

Proof. Since $V < 0$ on the compact set Z_0 , there exist an open neighborhood N of Z_0 and a constant $\eta > 0$ such that $V \leq -\eta$ on N . As $Z_0 \cap \text{bd}(E) = \emptyset$ and $E = \{U > 0\}$, every point of Z_0 has a neighborhood on which $U \leq 0$; by compactness, after shrinking N if necessary we may also assume

$$U \leq 0 \quad \text{on } N.$$

Hence

$$U + \varepsilon V \leq -\varepsilon\eta < 0 \quad \text{on } N$$

for every $\varepsilon > 0$.

Next choose pairwise disjoint neighborhoods I_j^- and I_j^+ of a_j and b_j , respectively, small enough that they are disjoint from N , that U and V are C^1 there, and that for some constants $c_j^\pm > 0$ one has

$$U'(x) \geq 2c_j^- \quad (x \in I_j^-), \quad U'(x) \leq -2c_j^+ \quad (x \in I_j^+).$$

Since V' is bounded on each $I_j^\pm$, for all sufficiently small $\varepsilon > 0$ the derivative of $U + \varepsilon V$ keeps the same sign on these neighborhoods. Thus $U + \varepsilon V$ is strictly increasing on each I_j^- and strictly decreasing on each I_j^+ . Because $U(a_j) = U(b_j) = 0$ and V is continuous at the boundary points, shrinking the neighborhoods if necessary we may ensure that for every sufficiently small $\varepsilon > 0$ the function $U + \varepsilon V$ changes sign across each $I_j^\pm$. Hence there are unique points $a_j(\varepsilon) \in I_j^-$ and $b_j(\varepsilon) \in I_j^+$ with

$$U(a_j(\varepsilon)) + \varepsilon V(a_j(\varepsilon)) = 0, \quad U(b_j(\varepsilon)) + \varepsilon V(b_j(\varepsilon)) = 0,$$

and necessarily $a_j(\varepsilon) \rightarrow a_j$ and $b_j(\varepsilon) \rightarrow b_j$ as $\varepsilon \rightarrow 0^+$. Since

$$U(a_j(\varepsilon)) - U(a_j) = -\varepsilon V(a_j(\varepsilon)),$$

and V is bounded near a_j , we have $a_j(\varepsilon) - a_j = O(\varepsilon)$; similarly $b_j(\varepsilon) - b_j = O(\varepsilon)$. Dividing by ε and using the differentiability of U at the boundary points together with the continuity of V there, we obtain

$$a'_j(0) = -\frac{V(a_j)}{U'(a_j)}, \quad b'_j(0) = -\frac{V(b_j)}{U'(b_j)}.$$

Choose $R > 0$ so large that $\overline{E} \cup Z_0 \subset (-R, R)$ and $U \leq -2\kappa$ on $\mathbb{R} \setminus [-R, R]$ for some $\kappa > 0$. By condition (ii),

$$M := \max\left(1, \sup_{\mathbb{R} \setminus E} V\right) < \infty.$$

Hence

$$U + \varepsilon V \leq -2\kappa + \varepsilon M < -\kappa < 0 \quad \text{on } \mathbb{R} \setminus [-R, R]$$

whenever $0 < \varepsilon < \kappa/M$.

Now let

$$K := [-R, R] \setminus \left(N \cup \bigcup_{j=1}^N (I_j^- \cup I_j^+) \cup E\right)$$

and

$$K_+ := \overline{E} \setminus \bigcup_{j=1}^N (I_j^- \cup I_j^+).$$

The compact set K is contained in $\mathbb{R} \setminus E$ and satisfies $U < 0$ there. Since U is continuous on $\mathbb{R} \setminus E$, it follows that $U \leq -\kappa'$ on K for some $\kappa' > 0$. Since $K \subset \mathbb{R} \setminus E$, the same bound $V \leq M$ holds on K , and therefore

$$U + \varepsilon V \leq -\kappa' + \varepsilon M < 0 \quad \text{on } K$$

whenever $0 < \varepsilon < \kappa'/M$.

On the other hand, K_+ is a compact subset of E . Since U is lower semicontinuous and strictly positive on E , it follows that $U \geq m > 0$ on K_+ for some $m > 0$. By condition (iv), V is bounded below on K_+ ; writing

$$M_+ := \max\left(1, -\inf_{K_+} V\right),$$

we have $V \geq -M_+$ there. Hence

$$U + \varepsilon V \geq m - \varepsilon M_+ > 0 \quad \text{on } K_+$$

whenever $0 < \varepsilon < m/M_+$. Thus, for small $\varepsilon > 0$, the only sign changes of $U + \varepsilon V$ occur inside the neighborhoods $I_j^\pm$, and hence

$$E_\varepsilon = \bigsqcup_{j=1}^N (a_j(\varepsilon), b_j(\varepsilon)).$$

Therefore

$$|E_\varepsilon| = \sum_{j=1}^N (b_j(\varepsilon) - a_j(\varepsilon)),$$

and differentiation at $\varepsilon = 0^+$ yields the formula. $\square$

Now let μ be a minimizing probability measure. For ease of reference, we isolate the hypotheses used for the rest of the section.

Standing hypotheses for Section 4.

(S1)

$$E_\mu := \{U_\mu > 0\} = \bigsqcup_{j=1}^N (a_j, b_j)$$

is a finite union of disjoint bounded open intervals, and U_μ is lower semicontinuous on $\mathbb{R}$ and continuous on $\mathbb{R} \setminus E_\mu$;

(S2) U_μ is C^1 on a neighborhood of $\text{bd}(E_\mu)$;

(S3)

$$U_\mu(a_j) = U_\mu(b_j) = 0, \quad U'_\mu(a_j) > 0, \quad U'_\mu(b_j) < 0;$$

(S4)

$$Z_0 := \{U_\mu = 0\} \setminus \text{bd}(E_\mu)$$

is compact.

Thus (S2)–(S3) are the boundary regularity and nondegeneracy assumptions needed to differentiate the moving endpoints, while (S4) isolates the interior zero set on which later multipliers may live. For logarithmic potentials of nonzero compactly supported positive measures, lower semicontinuity and the condition $U_\mu(x) < 0$ for $|x|$ sufficiently large are automatic. The additional standing assumptions therefore concern C^1 regularity near the boundary and compactness of Z_0 .

Define

$$\lambda_\partial := \sum_{a \in \text{bd } E_\mu} \frac{1}{|U'_\mu(a)|} \delta_a.$$

We refer to λ_∂ as the *boundary measure*. It records only the moving boundary points of E_μ ; interior points of Z_0 will be handled separately by the multiplier ξ .

Let $\mathcal{M}_{\text{reg}}$ denote the class of finite positive measures ν supported in $[-1, 1]$ such that:

(M1) U_ν is C^1 on a neighborhood of $\text{bd}(E_\mu)$;

(M2) U_ν is continuous on a neighborhood of Z_0 ;

(M3)

$$\sup_{\mathbb{R} \setminus E_\mu} U_\nu < \infty.$$

In other words, $\mathcal{M}_{\text{reg}}$ consists of those positive perturbation measures whose potentials are regular enough near the moving boundary and on the active zero set for Proposition 4.1 to apply. For such ν , condition (iv) of Proposition 4.1 is automatic: if $K \subset E_\mu$ is compact, then

$$M_K := \max\{|x - t| : x \in K, t \in [-1, 1]\} < \infty,$$

and therefore

$$U_\nu(x) = \int \log \frac{1}{|x - t|} d\nu(t) \geq -\nu([-1, 1]) \log M_K \quad (x \in K).$$

Lemma 4.2 (Mutual-energy symmetry). *Let α, β be finite positive measures with compact support in $\mathbb{R}$. Then*

$$\int U_\alpha d\beta = \int U_\beta d\alpha$$

as an identity in $(-\infty, +\infty]$. In particular, if one side is finite, then both sides are finite and equal.

Proof. Choose $R > 0$ so that $\text{supp } \alpha \cup \text{supp } \beta \subset [-R, R]$, and define

$$K_R(x, t) := \log \frac{2R}{|x - t|}.$$

Since $|x - t| \leq 2R$ on $[-R, R]^2$, one has $K_R \geq 0$ there. Tonelli's theorem therefore gives

$$\iint K_R(x, t) d\beta(x) d\alpha(t) = \iint K_R(x, t) d\alpha(t) d\beta(x).$$

Expanding K_R yields

$$\int U_\alpha d\beta + \alpha(\mathbb{R})\beta(\mathbb{R}) \log(2R) = \int U_\beta d\alpha + \alpha(\mathbb{R})\beta(\mathbb{R}) \log(2R),$$

and hence the claim. □

Corollary 4.3 (Boundary variation inequality). *Under the regularity assumptions above, let $\nu \in \mathcal{M}_{\text{reg}}$. If*

$$U_\nu < 0 \quad \text{on } Z_0,$$

then

$$\int U_\nu d\lambda_\partial \geq 0.$$

Equivalently,

$$\int U_{\lambda_\partial} d\nu \geq 0.$$

Proof. Apply Proposition 4.1 with $U = U_\mu$ and $V = U_\nu$. The compact-set lower bound displayed just above verifies condition (iv), and the hypothesis $U_\nu < 0$ on Z_0 rules out the creation of new positive components from the interior zero set, so

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0^+} L(\mu + \varepsilon\nu) = \int U_\nu d\lambda_\partial.$$

Since $L(c\rho) = L(\rho)$ for every $c > 0$, the perturbation

$$\rho_\varepsilon := \frac{\mu + \varepsilon\nu}{1 + \varepsilon\nu([-1, 1])}$$

is again an admissible probability measure and satisfies $L(\rho_\varepsilon) = L(\mu + \varepsilon\nu)$. Thus the minimality of μ implies that this right derivative is nonnegative. The equivalent formulation follows from Lemma 4.2. $\square$

The next step is to adjoin a multiplier supported on the active interior zero set. This yields the dual measure naturally associated with the regular variational problem.

Theorem 4.4 (Regular dual completion and dual positivity). *Assume the regularity hypotheses above. Then there exist a scalar $c \geq 0$ and a finite positive measure ξ supported on Z_0 , not both zero, such that, with*

$$\lambda := c\lambda_\partial + \xi,$$

one has

$$\int U_\nu d\lambda \geq 0 \quad \text{for every } \nu \in \mathcal{M}_{\text{reg}},$$

and

$$\int U_\mu d\lambda = \int U_\lambda d\mu = 0.$$

Moreover:

(a) if $J \subset [-1, 1]$ is a closed interval whose endpoints do not lie in $\text{bd}(E_\mu)$, then

$$\int_J U_\lambda(x) dx \geq 0;$$

(b) if $x \in [-1, 1] \setminus (Z_0 \cup \text{bd}(E_\mu))$, then

$$U_\lambda(x) \geq 0;$$

(c) in particular,

$$U_\lambda \geq 0 \quad \text{for Lebesgue-a.e. } x \in [-1, 1].$$

Proof. Because $\mathcal{M}_{\text{reg}}$ is closed under addition and multiplication by positive scalars, the set

$$\mathcal{C} := \left\{ (U_\nu|_{Z_0}, \int U_\nu d\lambda_\partial) : \nu \in \mathcal{M}_{\text{reg}} \right\} \subset C(Z_0) \times \mathbb{R}$$

is a convex cone containing $(0, 0)$. By Corollary 4.3, no element of $\mathcal{C}$ can have first component strictly negative on Z_0 and second component strictly negative. Thus

$$\mathcal{C} \cap \mathcal{O} = \emptyset, \quad \mathcal{O} := \{(f, s) \in C(Z_0) \times \mathbb{R} : f < 0 \text{ on } Z_0, s < 0\}.$$

Here $\mathcal{O}$ is an open convex cone. A standard separation theorem for a convex cone and a disjoint open convex set therefore gives a nonzero continuous linear functional on $C(Z_0) \times \mathbb{R}$ of the form

$$(f, s) \mapsto \int_{Z_0} f d\tilde{\xi} + cs$$

with $\tilde{\xi}$ a finite signed measure on Z_0 and $c \in \mathbb{R}$, such that it is nonnegative on $\mathcal{C}$ and negative on $\mathcal{O}$; since $(0, 0) \in \mathcal{C}$, the separation level has been normalized to 0. We claim that in fact $\tilde{\xi} \geq 0$ and $c \geq 0$. First, for every $\varepsilon > 0$ one has $(-\varepsilon, -1) \in \mathcal{O}$, so

$$-\varepsilon \tilde{\xi}(Z_0) - c < 0.$$

Letting $\varepsilon \rightarrow 0^+$ gives $c \geq 0$. Next, let $h \in C(Z_0)$ satisfy $h \geq 0$. For every $\delta, \varepsilon > 0$, the pair $-(h + \delta), -\varepsilon$ lies in $\mathcal{O}$, and hence

$$-\int_{Z_0} h d\tilde{\xi} - \delta \tilde{\xi}(Z_0) - \varepsilon c < 0.$$

Using $c \geq 0$ and letting first $\varepsilon \rightarrow 0^+$ and then $\delta \rightarrow 0^+$, we conclude that $\int_{Z_0} h d\tilde{\xi} \geq 0$ for every $h \geq 0$, hence $\tilde{\xi} \geq 0$.

Now set $\xi := \tilde{\xi}$ and

$$\lambda := c \lambda_{\partial} + \xi.$$

Then for every $\nu \in \mathcal{M}_{\text{reg}}$,

$$\int_{Z_0} U_{\nu} d\tilde{\xi} + c \int U_{\nu} d\lambda_{\partial} \geq 0,$$

which is exactly

$$\int U_{\nu} d\lambda \geq 0.$$

This proves the first asserted inequality. Since the separating functional is nonzero, c and ξ are not both zero.

Because λ is supported on $\text{bd}(E_{\mu}) \cup Z_0 \subset \{U_{\mu} = 0\}$, one has

$$\int U_{\mu} d\lambda = 0.$$

By Lemma 4.2,

$$\int U_{\lambda} d\mu = \int U_{\mu} d\lambda = 0.$$

This proves the energy identity.

We next derive the positivity consequences that are genuinely forced by the variational argument. Let $J \subset [-1, 1]$ be a closed interval whose endpoints do not lie in $\text{bd}(E_{\mu})$, and define

$$\nu := \mathbf{1}_J(x) dx.$$

Then ν is a finite positive measure supported in $[-1, 1]$, the potential U_{ν} is continuous on $\mathbb{R}$, and U_{ν} is C^1 near every boundary point of E_{μ} because the only singularities of U'_{ν} occur at the endpoints of J . Hence $\nu \in \mathcal{M}_{\text{reg}}$. Applying the first part of the theorem and then Lemma 4.2 gives

$$0 \leq \int U_{\nu} d\lambda = \int U_{\lambda} d\nu = \int_J U_{\lambda}(x) dx,$$

which proves (a).

Now let $x \in [-1, 1] \setminus (Z_0 \cup \text{bd}(E_\mu))$. Since λ is supported on $Z_0 \cup \text{bd}(E_\mu)$, the logarithmic potential U_λ is continuous on a neighborhood of x . Suppose for contradiction that $U_\lambda(x) < 0$. By continuity, there exists a closed interval $J \subset [-1, 1] \setminus (Z_0 \cup \text{bd}(E_\mu))$ containing x in its interior such that

$$U_\lambda < 0 \quad \text{on } J.$$

The endpoints of J lie outside $\text{bd}(E_\mu)$, so part (a) applies and gives

$$0 \leq \int_J U_\lambda(t) dt,$$

contradicting $U_\lambda < 0$ on J . Hence $U_\lambda(x) \geq 0$, proving (b).

Finally, logarithmic potentials of finite compactly supported positive measures are locally integrable. If the set $\{x \in [-1, 1] : U_\lambda(x) < 0\}$ had positive Lebesgue measure, then by the Lebesgue differentiation theorem there would exist a compact interval $J \subset [-1, 1]$ with endpoints outside the finite set $\text{bd}(E_\mu)$ such that

$$\int_J U_\lambda(x) dx < 0,$$

contradicting (a). Therefore $U_\lambda \geq 0$ for Lebesgue-a.e. $x \in [-1, 1]$, proving (c). $\square$

Corollary 4.5 (Absolute continuity on the active zero set implies the Euler–Lagrange vanishing). *Assume the hypotheses of Theorem 4.4, and suppose in addition that*

$$\mu|_{Z_0} \ll dx.$$

Then

$$U_\lambda = 0 \quad \text{for } \mu\text{-almost every } x \in [-1, 1].$$

Proof. By Theorem 4.4(b), one has

$$U_\lambda(x) \geq 0 \quad \text{for every } x \in [-1, 1] \setminus (Z_0 \cup \text{bd}(E_\mu)).$$

By Theorem 4.4(c) and the absolute continuity assumption, one also has $U_\lambda \geq 0$ for μ -almost every point of Z_0 .

It remains only to treat $\text{bd}(E_\mu)$. Since $\text{bd}(E_\mu)$ is finite, any positive μ -mass there would have to come from an atom; but an atom at a boundary point p would force $U_\mu(p) = +\infty$, contradicting $U_\mu(p) = 0$. Hence

$$\mu(\text{bd}(E_\mu)) = 0.$$

Therefore $U_\lambda \geq 0$ for μ -almost every point of $[-1, 1]$. Combining this with the identity

$$\int U_\lambda d\mu = 0$$

from Theorem 4.4 yields

$$U_\lambda = 0 \quad \text{for } \mu\text{-almost every } x \in [-1, 1],$$

as claimed. $\square$

Remark 4.6. This is the point at which the assumption $\mu|_{Z_0} \ll dx$ enters: it lets one pass from Lebesgue-a.e. positivity to μ -a.e. positivity on the active zero set. In the one-cut ansatz considered later, the mass on the flat part of $\{U_\mu = 0\}$ is described by an explicit density.

Remark 4.7. The theorem produces a dual measure

$$\lambda = c \lambda_\partial + \xi, \quad c \geq 0, \quad \text{supp } \xi \subset Z_0,$$

for which $\int U_\nu d\lambda \geq 0$ for every regular perturbation measure ν , together with pointwise positivity away from $Z_0 \cup \text{bd}(E_\mu)$ and averaged positivity on all of $[-1, 1]$. Corollary 4.5 shows how the additional hypothesis $\mu|_{Z_0} \ll dx$ turns this into the vanishing statement $U_\lambda = 0$ for μ -almost every point. Section 5 revisits the same principal-value equations from the transport side.

5 One-cut formulas and transport perturbations

We now record the one-cut formulas and then discuss the transport calculation, which yields the same principal-value equations from the variational side.

We begin with the basic differentiated form of the constant-potential ansatz.

Proposition 5.1 (Differentiated one-cut equations). *Let $a \in (-1, 1)$.*

(a) *Suppose*

$$\lambda = \zeta_L \delta_{x_L} + \zeta_R \delta_{x_R} + g(x) \mathbf{1}_{[a, 1]}(x) dx$$

with $x_L, x_R \notin [a, 1]$, where $g \in L^p(a, 1)$ for some $1 < p < 2$ and $g \in C_{\text{loc}}^\alpha(a, 1)$ for some $\alpha > 0$. If U_λ is constant on $(a, 1)$, then for every $x \in (a, 1)$,

$$\text{p. v.} \int_a^1 \frac{g(y)}{x - y} dy = -\frac{\zeta_L}{x - x_L} - \frac{\zeta_R}{x - x_R}.$$

(b) *Suppose*

$$\mu = A \delta_{-1} + f(x) \mathbf{1}_{[a, 1]}(x) dx,$$

where $f \in L^p(a, 1)$ for some $1 < p < 2$ and $f \in C_{\text{loc}}^\alpha(a, 1)$ for some $\alpha > 0$. If U_μ is constant on $(a, 1)$, then for every $x \in (a, 1)$,

$$\text{p. v.} \int_a^1 \frac{f(y)}{x - y} dy = -\frac{A}{x + 1}.$$

In either case, the singular integral equation is of Tricomi type.

Proof. We prove (a); the proof of (b) is identical. For $x \in (a, 1)$ one has

$$U_\lambda(x) = \zeta_L \log \frac{1}{|x - x_L|} + \zeta_R \log \frac{1}{|x - x_R|} + \int_a^1 \log \frac{1}{|x - y|} g(y) dy.$$

Differentiating the atomic terms gives

$$\frac{d}{dx} \left(\zeta_L \log \frac{1}{|x - x_L|} + \zeta_R \log \frac{1}{|x - x_R|} \right) = -\frac{\zeta_L}{x - x_L} - \frac{\zeta_R}{x - x_R}.$$

For the integral term, the standard differentiation formula for logarithmic potentials of L^p densities yields

$$\frac{d}{dx} \int_a^1 \log \frac{1}{|x - y|} g(y) dy = -\text{p. v.} \int_a^1 \frac{g(y)}{x - y} dy \quad \text{for a.e. } x \in (a, 1).$$

The local Hölder regularity of g upgrades the principal-value expression to a pointwise identity on $(a, 1)$. Since U_λ is constant on $(a, 1)$, its derivative vanishes there, giving the displayed equation for every $x \in (a, 1)$. $\square$

The next lemma records the two basic building blocks: the equilibrium density on $[a, 1]$ and the signed balayage correction associated with a point mass from $\mathbb{R} \setminus [a, 1]$. In the notation below, the actual balayage density is $-\beta_{a,t}$.

Lemma 5.2 (One-cut building blocks). *Let $a \in (-1, 1)$, and define*

$$R(z) := \sqrt{(z-a)(z-1)}, \quad \Phi_a(z) := \frac{2z - a - 1 + 2R(z)}{1-a},$$

where R is taken on the branch with $R(z) \sim z$ as $z \rightarrow \infty$. Also set

$$\omega_a(x) := \frac{1}{\pi \sqrt{(1-x)(x-a)}} \quad (x \in (a, 1)).$$

Then:

(a) For every real $z \notin [a, 1]$,

$$U_{\omega_a dx}(z) = \log \frac{4}{1-a} - \log |\Phi_a(z)|.$$

In particular,

$$U_{\omega_a dx}(x) \equiv \log \frac{4}{1-a} \quad (x \in [a, 1]).$$

(b) For $t \in \mathbb{R} \setminus [a, 1]$, define the signed density

$$\beta_{a,t}(x) := \frac{R(t)}{\pi(x-t)\sqrt{(1-x)(x-a)}} \quad (x \in (a, 1)).$$

Equivalently, the balayage of δ_t onto $[a, 1]$ has density $-\beta_{a,t}(x)$. Then for every real $z \notin [a, 1] \cup \{t\}$,

$$U_{\delta_t + \beta_{a,t} dx}(z) = \log \left| \frac{\Phi_a(z)\Phi_a(t) - 1}{\Phi_a(t)(\Phi_a(z) - \Phi_a(t))} \right|.$$

In particular,

$$U_{\delta_t + \beta_{a,t} dx}(x) \equiv -\log |\Phi_a(t)| \quad (x \in [a, 1]).$$

Proof. For $z \in \mathbb{C} \setminus [a, 1]$, set

$$F_a(z) := \int_a^1 \frac{\omega_a(x)}{z-x} dx.$$

By the standard one-cut Cauchy-transform formula,

$$F_a(z) = \frac{1}{R(z)}.$$

Likewise, for $t \notin [a, 1]$,

$$G_{a,t}(z) := \int_a^1 \frac{\beta_{a,t}(x)}{z-x} dx = \frac{R(t)}{\pi} \int_a^1 \frac{dx}{(x-t)(z-x)\sqrt{(1-x)(x-a)}}.$$

Using the partial-fraction identity

$$\frac{1}{(x-t)(z-x)} = \frac{1}{z-t} \left(\frac{1}{x-t} + \frac{1}{z-x} \right)$$

and the formula for F_a , one obtains

$$G_{a,t}(z) = \frac{-1 + R(t)/R(z)}{z - t}.$$

Also,

$$\frac{\Phi'_a(z)}{\Phi_a(z)} = \frac{1}{R(z)}, \quad \frac{d}{dz} \log \frac{\Phi_a(z)\Phi_a(t) - 1}{\Phi_a(z) - \Phi_a(t)} = -\frac{R(t)}{(z - t)R(z)}.$$

For real $z \notin [a, 1]$, differentiating the logarithmic potential of $\omega_a(x) dx$ gives

$$\frac{d}{dz} U_{\omega_a dx}(z) = -F_a(z) = -\frac{1}{R(z)} = -\frac{d}{dz} \log \Phi_a(z).$$

Thus the difference between the two sides is constant on each of the two connected components of $\mathbb{R} \setminus [a, 1]$. On both components, that constant is 0, because the difference tends to 0 as $z \rightarrow \pm\infty$. This gives the displayed formula for $U_{\omega_a dx}(z)$ when $z \in \mathbb{R} \setminus [a, 1]$.

For $x \in (a, 1)$, the boundary value of F_a satisfies

$$\Re F_{a,+}(x) = \Re \frac{1}{R_+(x)} = 0,$$

so

$$\text{p. v.} \int_a^1 \frac{\omega_a(y)}{x - y} dy = 0.$$

Hence $\frac{d}{dx} U_{\omega_a dx}(x) = 0$ on $(a, 1)$, and therefore $U_{\omega_a dx}$ is constant on $[a, 1]$. Letting $z \rightarrow 1^+$ in the exterior formula yields

$$U_{\omega_a dx}(x) \equiv \log \frac{4}{1 - a} \quad (x \in [a, 1]),$$

proving (a).

Similarly, for real $z \notin [a, 1] \cup \{t\}$,

$$\frac{d}{dz} U_{\delta_t + \beta_{a,t} dx}(z) = -\frac{1}{z - t} - G_{a,t}(z) = -\frac{R(t)}{(z - t)R(z)}.$$

By the displayed derivative identity for Φ_a , this equals

$$\frac{d}{dz} \log \left| \frac{\Phi_a(z)\Phi_a(t) - 1}{\Phi_a(t)(\Phi_a(z) - \Phi_a(t))} \right|.$$

Hence the difference between the two sides is constant on each connected component of $\mathbb{R} \setminus ([a, 1] \cup \{t\})$. On the two unbounded components, that constant is 0, because the difference tends to 0 as $z \rightarrow \pm\infty$.

For $x \in (a, 1)$, the boundary value of $G_{a,t}$ satisfies

$$\Re G_{a,t,+}(x) = \Re \frac{-1 + R(t)/R_+(x)}{x - t} = -\frac{1}{x - t},$$

so

$$\text{p. v.} \int_a^1 \frac{\beta_{a,t}(y)}{x - y} dy = -\frac{1}{x - t}.$$

Hence $\frac{d}{dx} U_{\delta_t + \beta_{a,t} dx}(x) = 0$ on $(a, 1)$, and therefore $U_{\delta_t + \beta_{a,t} dx}$ is constant on $[a, 1]$. Since $|\Phi_a(x)| = 1$ for $x \in [a, 1]$ and $\Phi_a(t) \in \mathbb{R} \setminus [-1, 1]$,

$$\left| \frac{\Phi_a(x)\Phi_a(t) - 1}{\Phi_a(t)(\Phi_a(x) - \Phi_a(t))} \right| = \frac{1}{|\Phi_a(t)|}.$$

Letting $z \rightarrow 1^+$ in the already established exterior formula therefore gives

$$U_{\delta_t + \beta_{a,t}} dx(x) \equiv -\log |\Phi_a(t)| \quad (x \in [a, 1]).$$

Finally, continuity of both sides at the endpoint of $[a, 1]$ adjacent to the bounded component between t and $[a, 1]$ shows that the remaining constant is also 0. This proves (b). $\square$

The Cauchy-transform identity from the proof also gives, by evaluation at $z = -1$,

$$\int_a^1 \frac{\omega_a(x)}{x+1} dx = -\frac{1}{R(-1)}.$$

Combining Lemma 5.2 with the differentiated equations gives explicit formulas for the two one-cut ansatzes: the primal measure μ and the dual measure λ .

Proposition 5.3 (Explicit formulas for the one-cut ansatz). *Let $a \in (-1, 1)$.*

(a) *Suppose*

$$\mu = A\delta_{-1} + f(x)\mathbf{1}_{[a,1]}(x) dx$$

is a probability measure, where $f \in L^p(a, 1)$ for some $1 < p < 2$ and $f \in C_{\text{loc}}^\alpha(a, 1)$ for some $\alpha > 0$, and suppose that U_μ is constant on $(a, 1)$. Then necessarily

$$f(x) = \omega_a(x) + A\beta_{a,-1}(x) = \frac{1}{\pi} \frac{x + (1 - A\sqrt{2(1+a)})}{(x+1)\sqrt{(1-x)(x-a)}}.$$

If in addition $U_\mu \equiv 0$ on $(a, 1)$, then

$$A = A(a) := \frac{\log \frac{4}{1-a}}{\log \frac{3+a+2\sqrt{2(1+a)}}{1-a}} = \frac{\log \frac{4}{1-a}}{2 \log \frac{\sqrt{2}+\sqrt{1+a}}{\sqrt{1-a}}}.$$

(b) *Suppose*

$$\lambda = \zeta_L \delta_{x_L} + \zeta_R \delta_{x_R} + g(x)\mathbf{1}_{[a,1]}(x) dx$$

with $x_L, x_R \notin [a, 1]$, where $g \in L^p(a, 1)$ for some $1 < p < 2$ and $g \in C_{\text{loc}}^\alpha(a, 1)$ for some $\alpha > 0$, and suppose that U_λ is constant on $(a, 1)$. Then necessarily

$$g(x) = c_0 \omega_a(x) + \zeta_L \beta_{a,x_L}(x) + \zeta_R \beta_{a,x_R}(x),$$

that is,

$$g(x) = \frac{1}{\pi \sqrt{(1-x)(x-a)}} \left(c_0 + \frac{\zeta_L R(x_L)}{x - x_L} + \frac{\zeta_R R(x_R)}{x - x_R} \right)$$

for some constant $c_0 \in \mathbb{R}$. If in addition $U_\lambda \equiv 0$ on $(a, 1)$, then

$$c_0 = \frac{\zeta_L \log |\Phi_a(x_L)| + \zeta_R \log |\Phi_a(x_R)|}{\log \frac{4}{1-a}}.$$

Proof. For (a), Proposition 5.1(b) shows that

$$\text{p. v.} \int_a^1 \frac{f(y) - A\beta_{a,-1}(y)}{x - y} dy = 0 \quad (x \in (a, 1)).$$

By the standard description of the kernel of the finite Hilbert transform on one interval, the difference

$$f - A\beta_{a,-1}$$

must be a scalar multiple of ω_a ; say

$$f = c\omega_a + A\beta_{a,-1}.$$

Since μ is a probability measure and

$$\int_a^1 \omega_a(x) dx = 1, \quad \int_a^1 \beta_{a,-1}(x) dx = R(-1) \int_a^1 \frac{\omega_a(x)}{x+1} dx = -1,$$

we obtain

$$1 = \mu([-1, 1]) = A + \int_a^1 f(x) dx = A + c - A = c,$$

so $c = 1$. This proves the displayed formula for f . Using $R(-1) = -\sqrt{2(1+a)}$ gives the explicit rational numerator.

If in addition $U_\mu \equiv 0$ on $(a, 1)$, then Lemma 5.2 gives

$$0 = \log \frac{4}{1-a} - A \log |\Phi_a(-1)|,$$

which is the stated formula for $A(a)$ because

$$|\Phi_a(-1)| = \frac{3+a+2\sqrt{2(1+a)}}{1-a}.$$

For (b), Proposition 5.1(a) gives

$$\text{p. v.} \int_a^1 \frac{g(y) - \zeta_L \beta_{a,x_L}(y) - \zeta_R \beta_{a,x_R}(y)}{x-y} dy = 0 \quad (x \in (a, 1)).$$

Again the kernel of the finite Hilbert transform on $(a, 1)$ is spanned by ω_a , so

$$g = c_0 \omega_a + \zeta_L \beta_{a,x_L} + \zeta_R \beta_{a,x_R}$$

for some $c_0 \in \mathbb{R}$. If in addition $U_\lambda \equiv 0$ on $(a, 1)$, then Lemma 5.2 yields

$$0 = c_0 \log \frac{4}{1-a} - \zeta_L \log |\Phi_a(x_L)| - \zeta_R \log |\Phi_a(x_R)|,$$

which is exactly the displayed formula for c_0 . □

Corollary 5.4 (Normalization and positivity for the primal one-cut density). *Fix $a \in (-1, 1)$ and $A \in \mathbb{R}$, and define*

$$f_{a,A}(x) := \omega_a(x) + A\beta_{a,-1}(x) = \frac{1}{\pi} \frac{x + (1 - A\sqrt{2(1+a)})}{(x+1)\sqrt{(1-x)(x-a)}}.$$

Then

$$\int_a^1 f_{a,A}(x) dx = 1 - A,$$

and

$$f_{a,A} \geq 0 \text{ almost everywhere on } (a, 1) \iff A \leq \sqrt{\frac{1+a}{2}}.$$

Proof. By the identities

$$\int_a^1 \omega_a(x) dx = 1, \quad \int_a^1 \beta_{a,-1}(x) dx = R(-1) \int_a^1 \frac{\omega_a(x)}{x+1} dx = -1,$$

we obtain

$$\int_a^1 f_{a,A}(x) dx = 1 - A.$$

The denominator in the displayed formula for $f_{a,A}$ is positive on $(a, 1)$, while the numerator is increasing in x . Hence $f_{a,A} \geq 0$ almost everywhere on $(a, 1)$ if and only if the numerator is nonnegative at $x = a$, namely if and only if

$$1 + a - A\sqrt{2(1+a)} \geq 0,$$

which is equivalent to

$$A \leq \sqrt{\frac{1+a}{2}}.$$

□

We now turn to transport perturbations. The discussion below provides another derivation of the same principal-value equations from the variational structure of the minimization problem. To obtain the principal-value equation on an active interval, one needs control of the transported potentials on that interval itself, not merely away from $\text{supp } \mu$. We therefore isolate the regularity assumptions needed for the transport argument.

Proposition 5.5 (Linearization of a transport perturbation away from the support). *Let μ be a finite positive measure supported in $[-1, 1]$, let $\phi : [-1, 1] \rightarrow \mathbb{R}$ be bounded, and for small ε define*

$$T_\varepsilon(x) := x + \varepsilon\phi(x), \quad \mu_\varepsilon := (T_\varepsilon)_\# \mu.$$

Assume that T_ε maps $[-1, 1]$ into itself and is increasing for all sufficiently small ε . Then for every compact set $K \subset \mathbb{R} \setminus \text{supp } \mu$, the map $(\varepsilon, x) \mapsto U_{\mu_\varepsilon}(x)$ is differentiable in ε on a neighborhood of $\{0\} \times K$, and

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} U_{\mu_\varepsilon}(x) = \int \frac{\phi(t)}{x-t} d\mu(t) \quad (x \in K).$$

In particular, the induced first-order perturbation of the potential is

$$V_\phi(x) := \int \frac{\phi(t)}{x-t} d\mu(t), \quad x \in \mathbb{R} \setminus \text{supp } \mu.$$

Proof. The identity for U_{μ_ε} is simply the definition of pushforward:

$$U_{\mu_\varepsilon}(x) = \int \log \frac{1}{|x-s|} d\mu_\varepsilon(s) = \int \log \frac{1}{|x-T_\varepsilon(t)|} d\mu(t).$$

Fix a compact set $K \subset \mathbb{R} \setminus \text{supp } \mu$. Since K and $\text{supp } \mu$ are disjoint, there exists $\delta > 0$ such that $|x-t| \geq 2\delta$ for all $x \in K$ and $t \in \text{supp } \mu$. For sufficiently small ε , one then has $|x-T_\varepsilon(t)| \geq \delta$ on $K \times \text{supp } \mu$. Hence the map

$$(\varepsilon, x, t) \mapsto \log \frac{1}{|x-T_\varepsilon(t)|}$$

is continuously differentiable in ε , uniformly on a neighborhood of $\{0\} \times K \times \text{supp } \mu$. Differentiating under the integral sign gives

$$\frac{d}{d\varepsilon} \log \frac{1}{|x - T_\varepsilon(t)|} = \frac{\phi(t)}{x - T_\varepsilon(t)}.$$

Evaluating at $\varepsilon = 0$ yields

$$\frac{d}{d\varepsilon} \Big|_{\varepsilon=0} U_{\mu_\varepsilon}(x) = \int \frac{\phi(t)}{x - t} d\mu(t),$$

uniformly for $x \in K$. □

Proposition 5.6 (Regular transport first variation). *Let μ be a minimizer satisfying the regularity assumptions of Proposition 4.1, and let $\phi : [-1, 1] \rightarrow \mathbb{R}$ be bounded. Assume that for all sufficiently small $\varepsilon > 0$ the map*

$$T_\varepsilon(x) = x + \varepsilon\phi(x)$$

maps $[-1, 1]$ into itself and is increasing, and write $\mu_\varepsilon := (T_\varepsilon)_\# \mu$. Assume in addition that there exist an open neighborhood Ω of $\text{bd}(E_\mu) \cup Z_0$, a bounded continuous function V_ϕ on Ω , and pairwise disjoint open intervals $I_j^-, I_j^+ \subset \Omega$ around the boundary points a_j, b_j such that:

(i) *one has the uniform first-order expansion*

$$\sup_{x \in \Omega} |U_{\mu_\varepsilon}(x) - U_\mu(x) - \varepsilon V_\phi(x)| = o(\varepsilon) \quad (\varepsilon \rightarrow 0^+);$$

(ii) *for each j , the function U_{μ_ε} is C^1 on $I_j^- \cup I_j^+$ for all sufficiently small $\varepsilon > 0$, and*

$$\sup_{x \in I_j^- \cup I_j^+} |U'_{\mu_\varepsilon}(x) - U'_\mu(x)| \rightarrow 0 \quad (\varepsilon \rightarrow 0^+);$$

(iii) $V_\phi < 0$ on Z_0 ;

(iv) *for every compact set $K \subset \mathbb{R} \setminus \Omega$, one has*

$$\sup_{x \in K} |U_{\mu_\varepsilon}(x) - U_\mu(x)| \rightarrow 0 \quad (\varepsilon \rightarrow 0^+);$$

(v) *for every compact set $K \subset E_\mu$, one has*

$$\inf_{x \in K} U_{\mu_\varepsilon}(x) > 0$$

for all sufficiently small $\varepsilon > 0$.

Then

$$\frac{d}{d\varepsilon} \Big|_{\varepsilon=0^+} L(\mu_\varepsilon) = \int V_\phi d\lambda_\partial.$$

If, in addition,

$$\sum_{a \in \text{bd}(E_\mu)} \int \frac{|\phi(t)|}{|a - t|} d\mu(t) < \infty,$$

and $V_\phi(x) = \int \frac{\phi(t)}{x - t} d\mu(t)$ for $x \in \text{bd}(E_\mu)$, then the right-hand side can be written as the absolutely convergent iterated integral

$$\int V_\phi d\lambda_\partial = \int \left(\int \frac{d\lambda_\partial(a)}{a - t} \right) \phi(t) d\mu(t).$$

Proof. Let N_0 be an open neighborhood of Z_0 with $\overline{N_0} \subset \Omega$ and $N_0 \cap \bigcup_j (I_j^- \cup I_j^+) = \emptyset$. Shrinking N_0 and the intervals $I_j^\pm$ if necessary, we may also assume that

$$\overline{I_j^- \cup I_j^+} \subset \Omega \quad (1 \leq j \leq N).$$

Shrinking N_0 further if necessary, we may assume both that

$$U_\mu \leq 0 \quad \text{on } N_0$$

and that

$$V_\phi \leq -\eta \quad \text{on } N_0$$

for some $\eta > 0$. By (i), the transported potentials satisfy

$$U_{\mu_\varepsilon}(x) = U_\mu(x) + \varepsilon V_\phi(x) + o(\varepsilon)$$

uniformly on $N_0 \cup \bigcup_j (I_j^- \cup I_j^+)$, and therefore $U_{\mu_\varepsilon} < 0$ on N_0 for all sufficiently small $\varepsilon > 0$.

By (ii) and the sign conditions $U'_\mu(a_j) > 0$, $U'_\mu(b_j) < 0$, the functions U_{μ_ε} are strictly increasing on I_j^- and strictly decreasing on I_j^+ for all sufficiently small $\varepsilon > 0$. The uniform expansion from (i) then gives unique zeroes $a_j(\varepsilon) \in I_j^-$ and $b_j(\varepsilon) \in I_j^+$ satisfying

$$U_{\mu_\varepsilon}(a_j(\varepsilon)) = U_{\mu_\varepsilon}(b_j(\varepsilon)) = 0.$$

Using the expansion from (i) at the boundary points exactly as in Proposition 4.1, one obtains

$$a'_j(0) = -\frac{V_\phi(a_j)}{U'_\mu(a_j)}, \quad b'_j(0) = -\frac{V_\phi(b_j)}{U'_\mu(b_j)}.$$

Set

$$\Omega_0 := N_0 \cup \bigcup_{j=1}^N (I_j^- \cup I_j^+).$$

Then Ω_0 is open, contains $Z_0 \cup \text{bd}(E_\mu)$, and satisfies $\overline{\Omega_0} \subset \Omega$. Next choose $R > 0$ so large that $\overline{E}_\mu \cup Z_0 \subset (-R, R)$ and $U_\mu < 0$ on $\mathbb{R} \setminus [-R, R]$. Since each μ_ε is again a probability measure supported in $[-1, 1]$, one also has $U_{\mu_\varepsilon} < 0$ on $\mathbb{R} \setminus [-R, R]$ for all sufficiently small $\varepsilon > 0$ after enlarging R once and for all. Let

$$K := [-R, R] \setminus (\Omega \cup E_\mu), \quad K_+ := \overline{E}_\mu \setminus \bigcup_{j=1}^N (I_j^- \cup I_j^+),$$

and define the compact “buffer” region

$$K_\Omega^- := [-R, R] \cap \overline{\Omega \setminus \Omega_0} \cap (\mathbb{R} \setminus E_\mu).$$

Then K is a compact subset of $\mathbb{R} \setminus \Omega$, and also of $\mathbb{R} \setminus E_\mu$. Since U_μ is continuous on $\mathbb{R} \setminus E_\mu$ and strictly negative on K , there exists $\kappa' > 0$ such that

$$U_\mu \leq -\kappa' \quad \text{on } K.$$

Hypothesis (iv) therefore gives $U_{\mu_\varepsilon} < 0$ on K for all sufficiently small $\varepsilon > 0$.

The set K_Ω^- is compact, and it is disjoint from $Z_0 \cup \text{bd}(E_\mu)$ because $Z_0 \cup \text{bd}(E_\mu) \subset \Omega_0$. Since also $K_\Omega^- \subset \mathbb{R} \setminus E_\mu$, continuity of U_μ on $\mathbb{R} \setminus E_\mu$ shows that $U_\mu < 0$ on K_Ω^- . By compactness, $U_\mu \leq -\kappa''$ on K_Ω^- for some $\kappa'' > 0$.

To pass from U_μ to U_{μ_ε} on K_Ω^- , split this set into

$$K_{\Omega, \text{in}}^- := K_\Omega^- \cap \Omega, \quad K_{\Omega, \text{out}}^- := K_\Omega^- \setminus \Omega.$$

On $K_{\Omega, \text{in}}^-$, hypothesis (i) gives uniform convergence $U_{\mu_\varepsilon} \rightarrow U_\mu$, because the error term is uniform on all of Ω and V_ϕ is bounded on Ω . On the compact set $K_{\Omega, \text{out}}^- \subset \mathbb{R} \setminus \Omega$, hypothesis (iv) likewise gives uniform convergence $U_{\mu_\varepsilon} \rightarrow U_\mu$. Hence $U_{\mu_\varepsilon} < 0$ on all of K_Ω^- for all sufficiently small $\varepsilon > 0$.

Likewise, K_+ is a compact subset of E_μ , so (v) gives $U_{\mu_\varepsilon} > 0$ on K_+ for all sufficiently small $\varepsilon > 0$. Thus, for small $\varepsilon > 0$, the only sign changes of U_{μ_ε} occur inside the neighborhoods $I_j^\pm$, and hence

$$\{U_{\mu_\varepsilon} > 0\} = \bigsqcup_{j=1}^N (a_j(\varepsilon), b_j(\varepsilon)).$$

Differentiating the total length now gives the claimed formula. The iterated-integral identity follows from the additional integrability hypothesis by Fubini's theorem. $\square$

Proposition 5.7 (Tangent transport). *Assume the hypotheses of Theorem 4.4. Let $J \subset (-1, 1)$ be an open interval. Assume that for every $\phi \in C_c^1(J)$ and all sufficiently small $|\varepsilon|$, the map*

$$T_\varepsilon(x) = x + \varepsilon \phi(x)$$

maps $[-1, 1]$ into itself and is increasing. Suppose further that for every such ϕ there exists a function V_ϕ , continuous on a neighborhood of $\text{bd}(E_\mu) \cup Z_0$, with

$$V_\phi = 0 \quad \text{on } Z_0,$$

and such that the two one-sided transport derivative formulas

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0^+} L((T_\varepsilon)_\# \mu) = \int V_\phi d\lambda_\partial, \quad \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0^+} L((T_{-\varepsilon})_\# \mu) = - \int V_\phi d\lambda_\partial$$

are valid. Then for every such ϕ ,

$$\int V_\phi d\lambda_\partial = 0 \quad \text{and} \quad \int V_\phi d\lambda = 0.$$

Proof. Fix such a test field ϕ . Since μ is minimizing, the two one-sided derivatives in the hypothesis are both nonnegative. Therefore

$$0 \leq \int V_\phi d\lambda_\partial \quad \text{and} \quad 0 \leq - \int V_\phi d\lambda_\partial,$$

so

$$\int V_\phi d\lambda_\partial = 0.$$

Since $\lambda = c \lambda_\partial + \xi$ and $\text{supp } \xi \subset Z_0$, the condition $V_\phi = 0$ on Z_0 gives

$$\int V_\phi d\lambda = c \int V_\phi d\lambda_\partial + \int V_\phi d\xi = 0.$$

$\square$

Corollary 5.8 (When transport yields the principal-value equation). *Assume the hypotheses of Proposition 5.7, and suppose in addition that there exists a function $H_\lambda \in L^1_{\text{loc}}(J)$ such that:*

(i) *for every $\phi \in C^1_c(J)$ one has*

$$\int V_\phi d\lambda = \int_J H_\lambda(t)\phi(t) d\mu(t);$$

(ii) *U_λ is absolutely continuous on every compact subinterval of J , and for almost every $t \in J$,*

$$H_\lambda(t) = \text{p. v.} \int \frac{d\lambda(a)}{a-t} = -(U_\lambda)'(t).$$

If μ is absolutely continuous on J with density positive almost everywhere on J , then

$$H_\lambda(t) = 0 \quad \text{for a.e. } t \in J,$$

and hence U_λ is constant on J .

Proof. By Proposition 5.7,

$$\int V_\phi d\lambda = 0 \quad \text{for every } \phi \in C^1_c(J).$$

Using (i), we obtain

$$\int_J H_\lambda(t)\phi(t) d\mu(t) = 0 \quad \text{for every } \phi \in C^1_c(J).$$

If $d\mu(t) = \rho(t) dt$ on J with $\rho > 0$ almost everywhere, this says

$$\int_J H_\lambda(t)\rho(t)\phi(t) dt = 0 \quad \text{for every } \phi \in C^1_c(J).$$

Thus the locally integrable function $H_\lambda\rho$ vanishes as a distribution on J , and hence vanishes almost everywhere. Since $\rho > 0$ almost everywhere, we conclude that $H_\lambda = 0$ almost everywhere on J . Property (ii) then shows that $(U_\lambda)' = 0$ almost everywhere on J and that U_λ is absolutely continuous on compact subintervals of J , hence U_λ is constant on J . $\square$

Taken together, Proposition 5.1, Proposition 5.3, and Corollary 5.8 exhibit the same singular-integral structure from two complementary directions. The direct one-cut ansatz begins with a constant-potential relation on an interval and differentiates it. The transport calculation begins with the variational characterization of a minimizer and, under the regular transport hypotheses from Proposition 5.7, leads to the same principal-value equation for the dual measure. In the one-cut setting, this is the mechanism behind the conclusion that U_λ is constant on the interval carrying the continuous part of μ .

References

- [1] *Forum discussion for Erdős Problem #1038*, <https://www.erdosproblems.com/forum/thread/1038>, accessed 2026-03-17.
- [2] P. Erdős, F. Herzog, and G. Piranian, *Metric properties of polynomials*, J. Analyse Math. **6** (1958), 125–148.
- [3] T. Tao, *Superlevel sets of logarithmic potentials*, <https://terrytao.wordpress.com/wp-content/uploads/2025/12/erdos-1038-2.pdf>, unpublished manuscript, draft dated 2025-12-27.